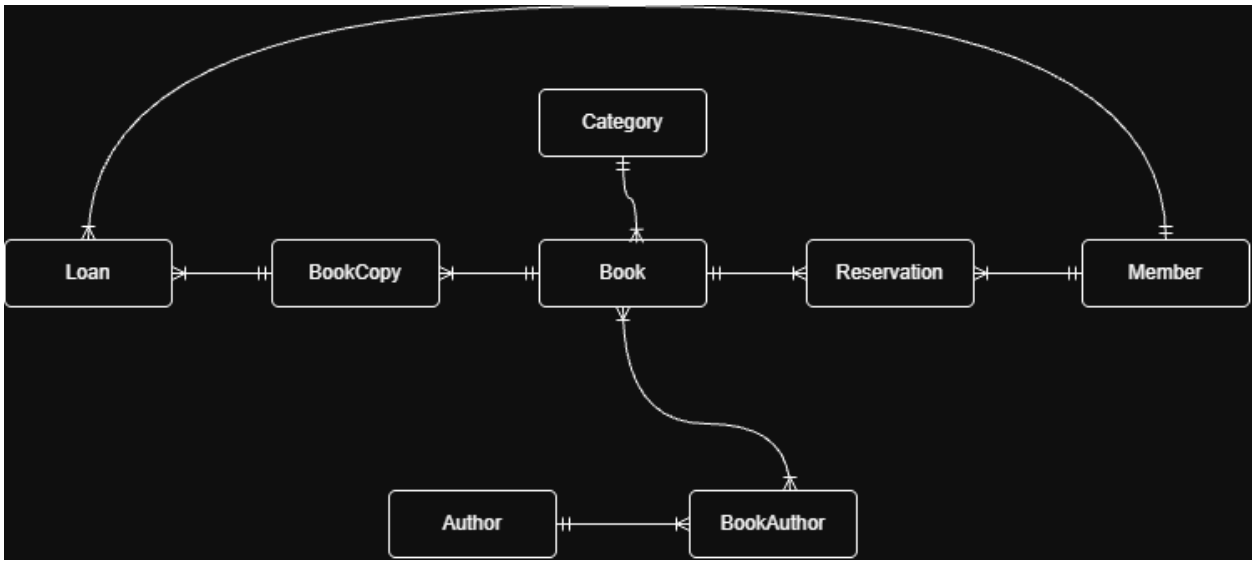
Lecture 5 - Homework 1

0. Domain: Library Management

1. Business Requirements

- Members can browse books
- A book may have multiple authors
- A book belongs to one category
- A title may have multiple physical copies
- Members borrow physical copies
- A member can have many loans
- A copy can only be in one active loan at a time
- Members can reserve a book if it is unavailable

2. ERD



BookAuthor is a junction entity used to resolve the many-to-many relationship between Book and Author.

3. Resource Design

Library Management application will have these API resources:

- /books
- /authors
- /categories
- /members
- /loans

- /reservations
- /books/{bookID}/copies. BookCopy is strongly dependent no book -> Nested resource is a better design than top-level.
- /book-authors

4. Endpoint list

- GET /books
- POST /books
- GET /books/{bookId}
- PATCH /books/{bookId}
- DELETE /books/{bookId}
- GET /books/{bookId}/copies
- POST /books/{bookId}/copies
- GET /books/{bookId}/copies/{copyId}
- GET /authors
- GET /authors/{authorId}
- GET /authors/{authorId}/books
- GET /categories
- GET /categories/{categoryId}
- GET /members/{memberId}
- GET /members/{memberId}/loans
- POST /loans
- GET /loans/{loanId}
- PATCH /loans/{loanId}
- POST /reservations
- GET /members/{memberId}/reservations